

Experiment – 11

Application to write data to SD card

AIM:

To develop an Android application that enables the user to store data in external storage (SD card) using file handling techniques.

ALGORITHM:

1. Start Application:

- *Open Android Studio and create a new project.*

2. Add Permissions:

- *Add write permission in AndroidManifest.xml.*
- *Request runtime permission if required.*

3. Design UI:

- *Add EditText for input.*
- *Add Button to save data.*

4. Initialize Components:

- *Link UI components using findViewById().*

5. Handle Button Click:

- *Set OnClickListener for save button.*

6. Check Storage:

- *Ensure external storage is available.*

7. Create File:

- *Create or access file in external storage.*

8. Write Data:

- *Use FileOutputStream to write data.*

9. Close Stream:

- *Close file after writing.*

10. **Display Message:**

- *Show Toast message after saving.*

11. **Handle Errors:**

- *Use try-catch for exceptions.*

CODE:

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<LinearLayout
```

```
    android:id="@+id/main"
```

```
    xmlns:android="http://schemas.android.com/apk/res/android"
```

```
    android:layout_width="match_parent"
```

```
    android:layout_height="match_parent"
```

```
    android:orientation="vertical"
```

```
    android:gravity="center"
```

```
    android:padding="20dp">
```

```
    <EditText
```

```
        android:id="@+id/editText"
```

```
        android:layout_width="match_parent"
```

```
        android:layout_height="wrap_content"
```

```
        android:hint="Enter text to save" />
```

```
    <Button
```

```
        android:id="@+id/btnSave"
```

```
        android:layout_width="match_parent"
```

```
        android:layout_height="wrap_content"
```

```
        android:text="Save to Storage"
```

```
        android:layout_marginTop="20dp"/>
```

```
</LinearLayout>
```

MainActivity.java

```
package com.example.sdcard;

import android.os.Bundle;
import android.os.Environment;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

import java.io.File;
import java.io.FileOutputStream;

public class MainActivity extends AppCompatActivity {

    EditText editText;
    Button btnSave;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        editText = findViewById(R.id.editText);
        btnSave = findViewById(R.id.btnSave);

        btnSave.setOnClickListener(v -> {

            String data = editText.getText().toString().trim();

            if (data.isEmpty()) {
                Toast.makeText(this, "Please enter text",
                    Toast.LENGTH_SHORT).show();
                return;
            }

            saveToStorage(data);
        });
    }
}
```

```

private void saveToStorage(String data) {
    try {
        File folder =
getExternalFilesDir(Environment.DIRECTORY_DOCUMENTS);

        if (folder != null && !folder.exists()) {
            folder.mkdirs();
        }

        File file = new File(folder, "sdcard_data.txt");

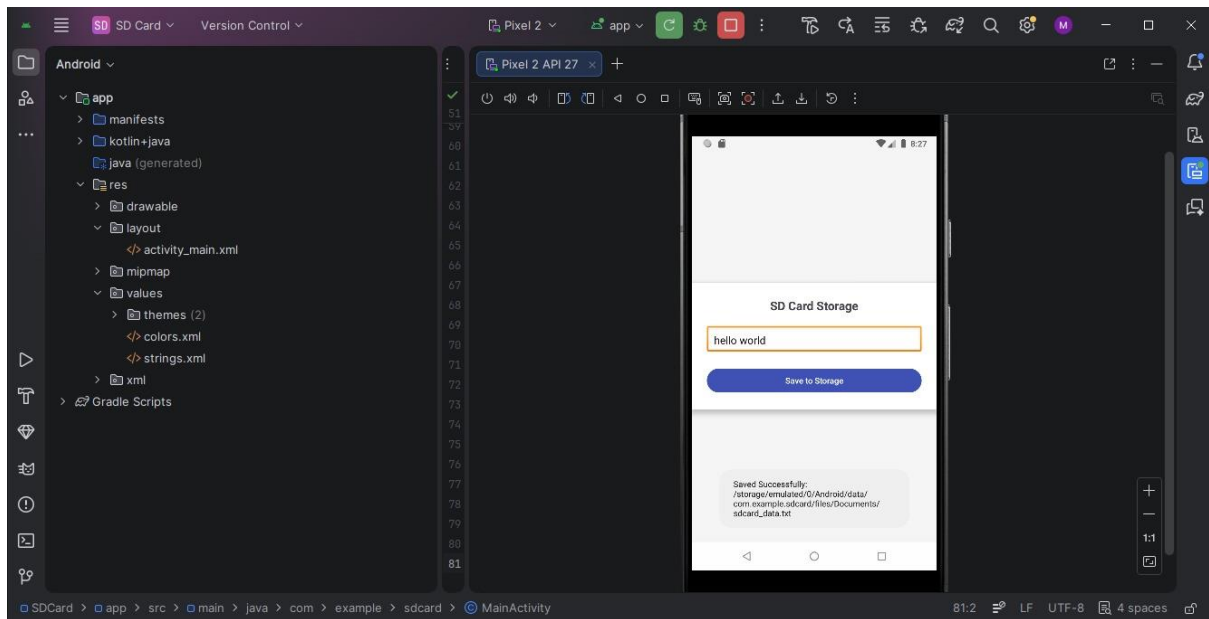
        FileOutputStream fos = new FileOutputStream(file);
        fos.write(data.getBytes());
        fos.close();

        Toast.makeText(this,
            "Saved Successfully",
            Toast.LENGTH_SHORT).show();

    } catch (Exception e) {
        Toast.makeText(this,
            "Error: " + e.getMessage(),
            Toast.LENGTH_SHORT).show();
    }
}
}

```

OUTPUT:



RESULT:

The Android application was successfully developed to write data into external storage (SD card).